Candidate-Bound Evidence for RTL-to-GDS Signoff: A Frozen-Source Evaluation

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Abstract. RTL2GDSAGI binds native verification evidence to the implementation selected for release. With frozen controller `af31b39f6782`, seven SKY130 designs spanning 16–1024 mapped cells complete the 21-stage verification contract twice. All seven pairs reproduce routed DEF, mapped and routed netlists, extracted parasitics, and date-normalized GDS; the 1,024-cell shift register uses one routing worker. Six constructed classes of stale or incomplete evidence are blocked in three consistency executions each. In the archived timing trajectories from snapshot `18d948815ac6`, 11/14 model proposals fail native checks, two lack completed target checks, and one passes. Frozen-source replay fails 14/14 timing states, but this comparison is confounded by a CTS-padding source change that reverses the archived pass. These proposal states were collected under iterative native feedback. A scripted baseline certifies spacing and PDN configurations without a model; 0/16 timing configurations certify at unchanged 3 ns. The nine live repair episodes are a recorded case study from snapshot `18d948815ac6`; live repair was not re-executed on the frozen source.

Keywords: RTL-to-GDS · physical verification · artifact provenance · electronic design automation

1 Introduction

OpenROAD and OpenLane provide open synthesis and physical-implementation infrastructure [1,2]. Release combines geometric, connectivity and timing evidence bound to the selected implementation and verification inputs. A zero-marker report may concern another GDS or omit required rules; repair can invalidate earlier evidence.

RTL2GDSAGI records artifact identity, consumed inputs, parser contracts and verdicts, with typed proposal authorization and native rechecking. We ask which constructed evidence failures are blocked, which recorded proposal states survive native checks, and which artifacts reproduce on a frozen controller. The contribution is this executable acceptance contract and its measured integration with existing native EDA algorithms.

2 Evidence Contract

2.1 Candidate identity and release checks

The controller executes characterization, lint, simulation, constraints, synthesis, synthesis equivalence, pre-layout timing, floorplan, power-network construction, placement, clock-tree synthesis (CTS), post-CTS timing, routing, extraction, post-route timing, GDS stream-out, DRC, LVS, antenna checking, routed equivalence, and signoff. Pre-layout timing is advisory; post-CTS timing gates hold slack, and post-route timing gates setup and hold. Release requires all declared hard obligations to have positive evidence.

The artifact ledger stores each output's producer, path, size, and SHA-256 hash. Directory hashing uses relative filenames and contents. Writing H for SHA-256, the candidate identity is

$$C = H(T, \{(k, H(A_k))\}_{k \in B}, P, V, H(K), H(I)) \,, \tag{1}$$

where T is the top, B the bound artifacts, P the recorded process-design kit (PDK) identity, V the tool identities, K the run configuration, and I the typed configuration state. P includes name, root, cell library, and corner. The layer map and verification decks are hashed members of B; the whole installed PDK is not content-hashed.

A gate records consumed and produced artifact hashes, report identity, parser contract, and accepted tool identity. Release revalidation rehashes files, checks lineage, and requires DRC and LVS to consume the selected GDS. Native KLayout arguments are checked against the selected input and deck bytes. The certification identifier commits to the candidate and gate records. Figure 1 summarizes the decision boundary.

Execution	Evidence obligation
RTL, constraints and native tools	Produce implementation and measured reports
Model proposal and typed authorization	Apply bounded changes and invalidate affected evidence
Native rechecking	Bind verdicts to consumed inputs and produced outputs
Candidate revalidation	Rehash selected artifacts and check gate lineage
Release	Certify only the complete declared contract

Fig. 1. Native evidence and content identity determine acceptance of a proposed implementation.

2.2 Native obligations and repair

DRC requires a usable report, the expected top, and the approved active-rule inventory under the exact deck. This evaluation uses 211 normalized categories; 213 raw declarations include two approved aliases. A zero-marker report with missing categories fails the inventory check. LVS requires agreement among process status, log verdict, and a native database with actual circuit cross-references, plus an approved deck containing executable compare. Stream-out requires nonempty geometry, resolved references, and mapped layers. The power distribution network (PDN) check requires supply nets, rails, straps, and vertical via connections.

Extraction compares logical net identities in Standard Parasitic Exchange Format (SPEF) and routed Design Exchange Format (DEF), rejects missing or duplicate nets, and requires complete parasitic sections. Native static timing analysis (STA) separately requires finite setup/hold slack and zero unannotated or partially unannotated drivers. Clock period, uncertainty, output load, RTL function, and verification decks are protected from physical-only repairs. The slack guardband $g = \texttt{sta.slack_guardband_ns} \geq 0$ is the minimum required slack: post-CTS requires hold WNS $\geq g$, and routed STA requires both setup and hold WNS $\geq g$. All evaluated runs use $g = 0$ ns. This accepts nonnegative slack without an additional robustness margin; the 0.25 ns clock uncertainty is a separate timing constraint. Neither a small positive slack nor exact repeat hashes establish margin against variation.

A typed change rolls execution back to its earliest affected stage and invalidates downstream evidence. The recorded live episodes configure three attempts per stage and 24 diagnosis attempts. These settings do not establish universal termination. The controller stops when a proposal would revisit a previously failed stage fingerprint, including its inputs and effective configuration. Formal equivalence checks both synthesis and routed netlists using EQY [3]. Scalar-net matching uses `splitnets on`; grouping related signals under `[collect *]` and `group *` preserves coupled-state proof structure. Fresh counter and shift-register controls prove equivalent, while a retained counter mutant with a forced flop fails equivalence.

3 Experimental Protocol

3.1 One frozen implementation

All primary implementation runs, fresh proposal executions, and repeated fault checks use source fingerprint `af31b39f6782`. It hashes sorted Python source paths and bytes. The frozen archive also retains non-Python resources. Current HEAD has this same implementation fingerprint; later Git commits change experiment scripts and manuscript material. The proposal-origin snapshot, `18d948815ac6`, is separate: it generated the archived live trajectories. Table 1 assigns each result to its execution and proposal origin.

The primary designs are a register, registered multiplexer, eight-bit ALU, counter, bidirectional shift register, four-entry FIFO, and 1,024-stage serial shift

Table 1. Source accounting. A source fingerprint differs from a Git commit identifier. Controlled ablations alter only the stated rendered padding calls.

Result	Source / origin
Seven primary design pairs	`af31b39f6782`
Fresh proposal and fault checks	`af31b39f6782`
24 scripted baseline states	Frozen native executor plus hashed stopping harness
Recorded model proposals / nine live episodes	`18d948815ac6`
Padding ablation	Frozen base; two explicit call interventions
Exploratory accumulator episodes	`af31b39f6782`

register. Each has a self-checking testbench. The PDK is SKY130A, Volare revision `bdc9412b3e46`, HD library, typical 25 °C and 1.8 V corner. The pinned container uses OpenROAD `41a51eaf4ca2` and KLayout 0.28.2; pre-layout OpenSTA reports 3.1.0. Baseline clocks are 10.0 ns, with 0.25 ns uncertainty and 0.05 pF output load.

Each design runs twice from RTL through signoff. The shift register uses one routing worker; its otherwise unchanged eight-worker pair is retained as an ablation. It uses two padding sites, 25% core utilization, 0.35 placement density, an 80% antenna prediction margin, eight repair iterations, and passive fillers. Final antenna checks use the original limits. All typed parameters, commands, tool/PDK identities, and candidate hashes are retained. Cell area comes from synthesis-library reports, die area from routed DEF, and timing from extracted parasitics. Concurrent-job runtime is an observation of this environment rather than isolated tool-performance measurement.

3.2 Coverage and fixed-proposal replay

The coverage unit is one of six author-constructed fault classes: stale same-top GDS, missing front-end-of-line (FEOL) rule categories, LVS without compare, missing PDN vias, missing layer map, and artifact substitution after a verified repair. Three executions per class check consistent behavior. Fixtures run native tools where applicable and exercise production parsers, structural checks, invocation checks, or gate lineage. Their selection against declared obligations makes this coverage testing, with no population false-pass-rate estimate.

The recorded proposals come from nine live Claude Sonnet 5 (Anthropic, `claude-sonnet-5`) episodes: three each for ALU spacing, register PDN construction, and ALU setup timing. Replay reconstructs complete effective configuration states, including characterization seeds, curated changes that preceded model calls, and cumulative authorized model changes. Successful reconstructions match independently retained archived final IRs for seven states, before adding the explicit routing-worker field absent from that schema. All 20 recon-

structions are consistent with archived defaults and logged changes; independent complete IR records for the other thirteen were not found in the inspected archive. All 20 states are then run independently from RTL on the frozen controller with one routing worker and further repair disabled. No new model calls generate these states. The supplementary record retains the original episodes, earlier measurements, reconstruction checks, and incomplete scale cases.

The hypothetical no-recheck gate promotes every authorized state. We report its failed-native-check fraction separately for each originating fixture class. The proposal sequence was collected under native feedback, so these fractions describe fixed states rather than the trajectory of an independently executed controller without rechecking.

3.3 Scripted baseline without a model

A predeclared sweep starts from the initial configurations of the three recorded fixtures. Spacing uses padding 0–3; PDN compares injected offset 50 μm with the schema default 2 μm. At fixed 3 ns, the timing grid crosses synthesis strategy DELAY0/DELAY1, density 0.45/0.55, padding 1/2, and native post-CTS setup repair off/on with a 0.05 ns target margin. The initial timing fixture uses AREA0 before these scripted changes. Two owner-declared 3.25/3.50 ns controls retain that initial configuration and change the clock requirement. The 0.2 delay fractions remain fixed, so absolute input/output delays scale from 0.60 ns to 0.65/0.70 ns. These controls are not an isolated clock-period sensitivity test or repairs of the 3 ns target.

All 24 states run from RTL with one routing worker and no model. A study-only harness stops at the first failed native hard gate; successful states require the frozen controller's complete certification. Native checks, constraints within each declared stratum, and verification decks are retained. The prior CLI pilot repeatedly rerouted unchanged IR because date-bearing GDS changed failed-input fingerprints. The study harness bounds this execution without changing the frozen controller. That pilot and the harness bootstrap failure remain excluded in the artifact. The supplement records the stopping policy and full candidate metrics.

4 Results

4.1 Implementation and exact repeat levels

Tables 2 and 3 report the first of each frozen pair. All 7 designs complete all 21 stages with zero routing, DRC, and antenna violations, positive LVS comparisons, both equivalence boundaries, authoritative simulation, and complete parasitic coverage. The largest design contains 1024 mapped cells.

Table 4 reports all seven pairs. Routed DEF, mapped and routed netlists, and SPEF match exactly for every pair. GDS matches for all seven after zeroing only the 24-byte BGNLIB and BGNSTR date payloads. Raw GDS matches in 1 pair; other raw differences retain the date fields. The one-worker shift pair matches

Table 2. Measured implementation results. Cell area is the synthesis-library area; die area comes from routed DEF. Timing uses extracted parasitics.

Design	Cells	Cell area (μm^2)	Die area (μm^2)	Setup slack (ns)	Hold slack (ns)
register	22	292.78	2393.17	6.954	1.204
mux4to1	48	480.46	3602.40	6.952	1.203
alu8	133	1018.48	6125.41	5.256	1.192
counter	47	416.65	3279.28	4.911	0.494
shift_reg	16	380.36	3091.92	6.892	0.554
fifo_buf	109	1721.65	9169.02	6.396	0.273
shift1024	1024	25624.58	115705.42	5.553	0.049

Table 3. Native verification and execution outcomes. Every design passed all 21 stages, both equivalence boundaries, LVS, and its authoritative testbench.

Design	Route/DRC violations	Antenna net/pin	SPEF nets	GDS (KiB)	Runtime (s)
register	0/0	0/0	35	131.8	31.7
mux4to1	0/0	0/0	87	161.6	45.8
alu8	0/0	0/0	157	497.2	73.1
counter	0/0	0/0	62	238.4	36.5
shift_reg	0/0	0/0	32	117.4	32.4
fifo_buf	0/0	0/0	126	486.6	54.4
shift1024	0/0	0/0	1210	3309.2	145.6

while the earlier eight-worker pair differs in DEF/GDS via variants, with identical netlists and SPEF. The retained pinned-binary query confirms the native one-thread default when no explicit thread command is emitted. This identifies worker-count sensitivity in the observed pairs, without isolating a particular native race.

Table 4. Seven frozen primary pairs. Date normalization zeros only BGNLIB/BGNSTR payloads; all other GDS bytes remain unchanged. Both netlists also match in every pair.

Design	Routed DEF	SPEF	Raw GDS	Dates zeroed
register	Same	Same	Same	Same
mux4to1	Same	Same	Different	Same
alu8	Same	Same	Different	Same
counter	Same	Same	Different	Same
shift_reg	Same	Same	Different	Same
fifo_buf	Same	Same	Different	Same
shift1024	Same	Same	Different	Same

A supplemental implementation fingerprint binds date-normalized GDS, RTL, routed netlist, SPEF, constraints, decks, layer map, and PDK name, library and corner. It matches for all seven primary pairs. It is a comparison identity; native release and gate identities continue to bind exact bytes, including reports and path-bearing characterization records. Normalization does not transfer a prior verdict to a changed artifact.

Post-hoc eligibility at $g = 50$ ps is 6/7 primary designs (12/14 runs), applying post-CTS hold and routed setup/hold. Both shift1024 repeats fail routed hold at 49.477 ps; the archived +1.586 ps repair fails setup. Other gate outcomes stay fixed; no new certificates are issued. The supplement reports all checkpoint values.

4.2 Fault coverage and native checker scope

Every constructed class is blocked in all three executions (Table 5). A native zero-marker result on stale geometry fails selected-input binding. Disabling FEOL produces an incomplete inventory; a fabricated match lacks circuit cross-references. Removed vias fail supply connectivity, omitted mapping fails layer checks, and substituted post-repair GDS invalidates cached gate lineage.

Table 5. Coverage of six constructed fault classes on the frozen controller. Three executions per class establish consistent fixture behavior, not independent samples.

Constructed class	Blocking obligation	Blocks
Stale same-top GDS	Selected input hash and invocation binding	3/3
Missing FEOL rules	Approved deck and active category inventory	3/3
LVS without compare	Executable comparison and circuit cross-references	3/3
Missing PDN vias	Structural supply and vertical connection checks	3/3
Missing layer map	Bound map and routed-layer geometry checks	3/3
Artifact swap after repair	Cached gate lineage and artifact rehashing	3/3

The OpenLane follow-up audits actual pinned stage/checker interfaces using paired controls. Table 6 reports the obligations within the remit of those interfaces. The previous modern-deck wrapper omitted runtime flags, leaving its rules inactive; it is excluded as an active-DRC baseline. Legacy native configuration and modern explicitly configured controls exercise the intended geometry rules. Public pinned source matches the retained wrapper, and PDK options are recorded. The modern adapter is explicitly identified. Modern configured controls report metal/poly/off-grid markers 3/1/4 while paired clean inputs remain clear; the legacy deck omits the tested poly rule. Reporting markers and enforcing a release decision are separate obligations. This study executes isolated

interfaces, without estimating full-flow OpenLane detection performance. The supplement contains the control counts, options, commands, and interface-remit qualifications.

Table 6. Obligation remit of the executed OpenLane interfaces and configured native controls. These are stage/checker observations, not a complete-flow detection comparison.

Obligation	Checker remit	Executed scope
Configured geometric marker reporting	Supported	Positive/clean controls; deck and options specified
Missing PDN via connectivity	Supported	Native supply checks with untouched DEF controls
Selected candidate and cached lineage	Not designed for	No release-manifest interface in these checker calls
Approved deck / active inventory	Not designed for	Rule execution differs from inventory approval
Retained KLayout no-compare fixture via Netgen	Not assessed	Report formats are incompatible
Mapping provenance	Not designed for	Required map option does not establish selected-map identity

4.3 Replay by originating class

Table 7 preserves both the original observations and fresh frozen-source outcomes. In the archived timing trajectories, 11 of 14 proposals fail native checks (78.6%), while 2 lack a completed required target check. Spacing and PDN each have zero failed proposals out of three. Thus the earlier aggregate 11/20 failure count comes entirely from timing proposals.

Table 7. Proposal replay by originating fixture class. A hypothetical gate accepting each proposal without rechecks would promote every row state. Failed means a native hard check failed; missing evidence is separate. Frozen outcomes are separate native executions of the same proposal states.

Class	Proposals	Archived failed	Archived missing	Frozen failed	Frozen certified
Spacing	3	0/3	0/3	0/3	3/3
PDN	3	0/3	0/3	0/3	3/3
Timing	14	11/14	2/14	14/14	0/14

On the frozen controller, spacing and PDN states each certify in three of three executions. All 14 timing states fail native hard checks (100.0%): thirteen fail routed STA and one fails placement. These are intermediate proposals in an iterative timing search, where failed attempts are expected. The replay supports retaining native checks as acceptance evidence; it supplies neither a general model-failure rate nor an end-to-end agent-superiority estimate.

Excluding `timing_1_proposal2`, the archived +1.586 ps state whose sign change is explained by CTS padding, leaves thirteen timing states. Twelve fail setup STA and one fails placement (DPL-0036), so a timing result is unavailable for that placement-failed state. The full fourteen-state denominator remains in Table 7.

4.4 Explaining the narrow timing repair

The archived repaired ALU configuration passes at +1.586 ps under its recorded source but fails at −0.654 ps under the frozen controller, at the same 3 ns period. We test CTS-padding reapplication, routing-padding reapplication, and routing workers in a $2 \times 2 \times 2$ factorial ablation. Placement padding stays at one site; clock, uncertainty, load, RTL, and PDK remain fixed. The supplement contains the full eight-run matrix.

Both worker counts give the same slack in every padding configuration. Reapplying CTS padding changes the archived-style result from +1.586 ps to −0.654 ps. Keeping routing padding while removing the CTS call gives +9.868 ps. Synthesis netlist and placed DEF hashes match before the intervention; CTS DEF and downstream artifacts differ, including legalized cell and hold-buffer positions. This isolates the CTS padding call's effect in these executions. The frozen controller retains that call and blocks the narrow configuration. The earlier pass therefore belongs to its recorded snapshot and is not presented as a repair reproducible on HEAD.

4.5 Results of the scripted baseline

Table 8 summarizes all declared strata; the retained `scripted_baseline_metrics.csv` gives all 24 candidate rows. Padding 0 fails DRC; padding 1, 2 and 3 each complete certification. Returning the PDN offset to 2 μm also certifies without the model's ring proposal. These fixtures therefore supply no evidence that an LLM is needed for spacing or PDN repair. They are manually selected finite configurations, not a population success rate or an autonomous diagnosis benchmark.

At unchanged 3 ns, 0/16 timing configurations certify; the best routed setup WNS is -0.000654 ns. All eight setup-repair off/on pairs have identical CTS DEF, routed DEF, SPEF and setup/hold slack. Instrumented CTS replay confirms all eight setup calls execute with pre-repair setup WNS 194.356–274.527 ps, above their 50 ps target under wire-load fallback, leaving no setup violation to repair at that point. The sixteen declared configurations therefore include eight outcome-equivalent toggle pairs; this is not a general result about setup-repair

Table 8. Finite scripted baseline through the same frozen native contract. Counts are configurations, not independent repair episodes. Slower-clock controls change the requirement and are excluded from 3 ns repair success. Best setup is the maximum over all candidates reaching routed STA, including later hard failures. The spacing maximum is DRC-failed padding 0.

Fixture	Scripted choices	Certified	Best setup all STA (ns)
Spacing	Padding 0–3 sites	3/4	+5.304609
PDN	Offset 50 or 2 μm	1/2	+6.953344
Timing, 3 ns	Strategy/density/padding/CTS grid	0/16	-0.000654
Slower-clock controls	3.25 or 3.50 ns	1/2	+0.060743

effectiveness. The 3.25 ns control fails at −0.141952 ns setup WNS; the 3.50 ns control certifies at +0.060743 ns. These are separate requirements, not repairs of the original target. The grid is not an exhaustive optimizer and does not establish incremental model value on timing. Model-generated proposals remain one possible input to the evidence contract; their unsupported states still require rejection.

5 Related Work

OpenROAD and OpenLane provide implementation infrastructure [1,2]. METRICS2.1 structures flow measurement and parameter tuning [4], and PROV represents provenance relationships [5]. Nix supports fixed-output hashing and experimental content-addressed derivations; Bazel maps action hashes to cached results and stores outputs by content [6,7]. SLSA build provenance relates artifact digests to build definitions and builder identities [8]. These mechanisms can carry domain-specific checks as declared actions and policies. This evaluation specializes release acceptance to native EDA invocation, report structure, approved inputs, and cached lineage; it does not evaluate Nix/Bazel execution or SLSA conformance.

ChatEDA studies model-driven EDA task planning, script generation, and execution [9]. ORFS-agent optimizes OpenROAD-flow-scripts parameters [10]; MCP4EDA connects model interaction to RTL-to-GDS execution and backend synthesis feedback [11]. FlowFixer-EDA uses log extraction, patch validation, and native tool feedback for diagnosis and repair [12]. Its patch validation already demonstrates the importance of rechecking. Here, the specific object of study is the release contract binding the selected GDS, approved verification inputs, native report structure, and cached gate lineage. EDAid uses multiple model-controlled agents for EDA workflow execution [13]. Ghose et al. study executable OpenROAD changes and measured quality of results [14]. Ghose et al.'s newer ICCAD author manuscript also records patch/snapshot/proof manifests

and renewed checks. Their manifest and rechecking mechanisms overlap with generic manifest-based revalidation; we claim no priority for that mechanism.

ChiBench supplies RTL designs for EDA-tool testing [15]. DRVerify documents physical-verification runset testing with known legal and violating structures [16]. That runset-testing rationale motivates checking rule execution and intended coverage in addition to marker counts. The present contribution combines constructed native/ledger faults, candidate lineage, fresh recorded-proposal checks, and artifact-level repeat evidence. Comparable contracts in other controllers are not excluded by this study.

6 Scope and Conclusion

The evidence establishes six-class coverage, class-specific proposal outcomes, and exact repeat levels on one frozen implementation. All seven primary pairs reproduce their routed and date-normalized artifacts. The replay's failures are confined to the timing fixture, and the padding experiment explains why its earlier marginal pass does not transfer to HEAD. The small synthetic circuits provide no population repair rate or industrial-scale qualification. The local contract uses a single typical corner and excludes multi-corner timing, on-chip variation, power integrity, and foundry qualification. Hashing also assumes trusted tools, controller, host, and PDK. Identity and approved inventory do not prove completeness of foundry rules or immunity to forged records.

The nine live episodes remain an archived case study on `18d948815ac6`; their 7/9 certifications are not a frozen-source repair rate. The separate exploratory accumulator cohort certifies 0 of 4: two policy escalations and two credit-interrupted episodes with unknown eventual outcomes. The largest clean design is the synthetic 1,024-cell shift register. The supplement retains earlier observations, failed-fingerprint proposals and incomplete scale probes. Logs retain updates and final artifacts; rollback can overwrite intermediate files. The regression record is 955 passed, 15 skipped, zero failures/errors.

Artifacts and AI assistance

Repository: `https://github.com/xp4t/rtl2gdsagi`. The versioned artifact at `rtl2gdsagi-paper-sensitivity-20261013` contains this LNCS manuscript, an Overleaf ZIP, `supplement.tex`, source snapshots, native reports, and table-generation/verification scripts. Codex assisted writing and three internal review passes in the same automated workflow. These are not independent external peer reviews.

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